

Abstract

The rapid expansion of video streaming applications, driven by live broadcasting, short-form content, and interactive media, has heightened the demand for adaptive bitrate (ABR) algorithms and congestion control mechanisms capable of balancing latency minimization, Quality of Experience (QoE) optimization, and dynamic network adaptation. This survey critically examines 15 foundational studies to dissect advancements across four domains: AI-driven adaptation, protocol-layer innovations, QoE-aware buffer management, and edge computing integrated with Non-Orthogonal Multiple Access (NOMA). The analysis reveals that machine learning techniques, such as edge-assisted models and multi-agent reinforcement learning (MARL), significantly enhance decision-making under volatile network conditions. For instance, edge computing with supervised learning reduces latency by 23%, while MARL frameworks improve QoE by 18% through expert-guided bitrate coordination. However, scalability challenges persist in multi-user environments, and computational overhead limits deployment on resource-constrained devices. Protocol innovations, such as deadline-aware congestion control, achieve 52% lower latency but struggle with cross-traffic volatility, underscoring the need for hybrid heuristic-reinforcement learning approaches. QoE optimization remains fragmented: predictive methods like GRU-based bandwidth forecasting reduce rebuffering by 41% but falter during sudden network dips, whereas reactive strategies stabilize playback at the cost of resolution quality. Theoretical frameworks expose critical gaps, such as the neglect of real-world interference in NOMA systems, which degrades QoE by 12% under dense network conditions. This survey proposes three transformative pathways: multi-objective MARL unifying bitrate fairness and physical-layer constraints, a hybrid QoE index balancing perceptual and technical metrics, and in-network NOMA architectures mitigating imperfect signal decoding. These contributions bridge AI-driven adaptability with protocol-layer pragmatism, offering a roadmap for scalable, human-centric streaming systems. The findings advocate for cross-layer benchmarking standards and industry-academia collaboration to accelerate deployment, positioning adaptive streaming for 6G-era demands.

1. Introduction

The explosive growth of video streaming, driven by diverse applications such as live broadcasting, short-form content platforms, and interactive media, has intensified the need for robust adaptive bitrate (ABR) algorithms and congestion control mechanisms. These systems must balance competing objectives: minimizing latency, maximizing Quality of Experience (QoE), and adapting to volatile network conditions. This survey synthesizes 15 pivotal works [1]–[15] to dissect advancements across four interconnected research domains—AI-driven adaptation, protocol-layer innovations, QoE-aware buffer management, and edge computing/NOMA integration—supported by theoretical frameworks that critique or validate experimental approaches.

AI/ML-Driven Adaptation: From Edge to Multi-Agent Systems

AI has emerged as a cornerstone for adaptive streaming, with techniques ranging from edge-assisted machine learning (ML) to decentralized reinforcement learning (RL). ECAS-ML [1] pioneers edge computing-assisted ML to offload bitrate decision-making, reducing latency by 23% compared to client-only ABR algorithms. However, its centralized edge dependency contrasts with multi-agent RL (MARL) frameworks like [2], which distribute adaptation across short video segments using expert-guided RL

agents. By integrating buffer and throughput heuristics, [2] achieves 18% higher QoE than traditional RL, though scalability in multi-user scenarios remains unaddressed [10]. Meanwhile, GRU-based bandwidth prediction [6] combines buffer dynamics with recurrent neural networks to forecast network capacity, reducing playback stalls by 41% but introducing computational overhead—a limitation partially mitigated by edge offloading in [1]. RL's versatility extends to TCP congestion control, where [4] employs deep RL (DRL) to optimize congestion windows dynamically, reducing retransmission timeouts by 34% in congested networks. However, hybrid heuristic-RL policies [13] critique purely data-driven methods for instability, advocating rule-based safeguards to ensure reliability.

Protocol Innovations: Low-Latency and Scalability Trade-offs

Protocol-layer advancements prioritize latency reduction and QoE stability but face inherent trade-offs. The ASViS protocol [3] introduces scalable video coding (SVC)-aware ABR logic, optimizing layer prioritization based on buffer health to outperform MPEG-DASH by 29% in PSNR. Yet, its reliance on stable throughput estimations conflicts with [8]'s dynamic bitrate analysis, which employs sliding-window HTTP adaptation to handle volatile networks, achieving 15% higher bitrate consistency. For ultra-low-latency streaming, SQP [7] redesigns congestion control around frame deadlines, slashing 95th percentile latency by 52% compared to Google's GCC. However, RFC 8867 [11]'s benchmarking reveals SQP's susceptibility to cross-traffic spikes, underscoring the need for hybrid approaches like [13]'s heuristic-RL fusion. In-network computing [14] further redefines protocol design by offloading adaptation logic to network middleboxes, reducing client overhead by 38%—a promising but underexplored direction that challenges traditional client-centric models [3], [7].

QoE Optimization: Predictive vs. Reactive Strategies

QoE-centric research bifurcates into predictive and reactive paradigms. GRU-based prediction [6] preempts congestion by forecasting bandwidth 5 seconds ahead, enabling proactive bitrate switching. Conversely, [9]'s buffer-focused algorithm reacts to sudden bandwidth drops by prioritizing chunk completion deadlines, sacrificing bitrate consistency for stability. Theoretical frameworks like [15] critique both approaches, emphasizing perceived quality metrics over technical indicators—a gap partially addressed by [5]'s real-world RL adaptation, which optimizes for subjective user ratings. Deadline-aware transmission [12] further bridges theory and practice, proposing chunk scheduling algorithms that align with human QoE sensitivity, though scalability in live streaming remains untested [14].

Edge and NOMA: Scalability vs. Imperfections

Edge computing and Non-Orthogonal Multiple Access (NOMA) introduce scalable multi-user solutions but grapple with practical imperfections. [10] integrates adaptive streaming with NOMA edge caching, demonstrating 20% higher spectral efficiency than OMA systems. However, imperfect successive interference cancellation (SIC) degrades performance by 12% in dense networks—a challenge unaddressed by existing ABR methods [3], [8]. ECAS-ML [1] mitigates edge limitations through ML-

driven resource allocation, yet its single-edge architecture conflicts with [14]’s distributed in-network computing model.

Theoretical Foundations: Benchmarking and Hybrid Models

Theoretical works provide critical evaluation frameworks. RFC 8867 [11] standardizes congestion control testing, exposing vulnerabilities in AI-driven [4] and protocol-layer [7] methods under real-world traffic variability. [12]’s deadline-aware model formalizes latency-QoE trade-offs, validating SQP’s [7] deadline prioritization but highlighting oversights in ASViS’s [3] buffer-centric approach. Hybrid heuristic-RL [13] and in-network computing [14] models further bridge simulation-to-reality gaps, advocating for layered optimization across network layers.

This survey systematically contrasts these paradigms, leveraging all 15 works [1]–[15] to identify synergies, critique limitations, and propose future directions grounded in empirical and theoretical insights.

2. Overview: The Evolution of Adaptive Streaming in the AI-Network Co-Design Era

The quest for seamless video streaming—spanning ultra-low-latency live broadcasts, binge-worthy long-form content, and viral short-form clips—has catalyzed a paradigm shift in adaptive streaming research. No longer confined to client-side heuristics or isolated protocol tweaks, modern solutions demand cross-layer orchestration between AI-driven intelligence, network protocol innovations, and human-centric quality metrics. This survey synthesizes 15 seminal works [1]–[15] to map this evolution, revealing four interconnected revolutions reshaping the field: 1) AI’s ascent from edge to multi-agent systems, 2) protocol-layer breakthroughs in latency-scalability trade-offs, 3) QoE optimization’s predictive-reactive duality, and 4) edge-NOMA synergies battling real-world imperfections. Below, we dissect these revolutions through a unified lens, exposing gaps and proposing pathways for 6G-ready streaming systems.

2.1 The AI Renaissance: From Edge Intelligence to Swarm Coordination

AI has transcended its role as a mere "optimization tool" to become the architectural backbone of adaptive streaming. Early edge-assisted models like ECAS-ML [1] demonstrated the power of offloading LSTM-based predictions to edge nodes, slashing client-side latency by 23% while achieving a 3.65 QoE score—unmatched by client-only ABR. Yet, its centralized edge dependency clashed with the distributed

ethos of multi-agent RL (MARL) frameworks like Incendio [2], which deploy expert-guided RL agents to coordinate bitrates across short video segments. By integrating buffer thresholds and throughput heuristics, [2] achieved 53% higher utility than traditional RL, but at the cost of scalability in dense NOMA networks [10]. Meanwhile, GRU-based predictors [6] and DRL-TCP controllers [4] showcased AI's versatility, reducing stalls by 41% and retransmission timeouts by 34%, respectively. However, these isolated advances expose a critical void: no framework unifies edge intelligence, multi-agent coordination, and physical-layer constraints (e.g., NOMA interference). The result? A fragmented ecosystem where edge offloading [1] conflicts with in-network computing [14], and MARL's bitrate fairness [2] ignores deadline-critical protocols like SQP [7].

2.2 Protocol Wars: Scalability vs. Realism

The protocol layer has become a battleground for competing visions of low-latency streaming. SVC-aware ABR protocols like ASViS [3] prioritize perceptual quality through layer switching, outperforming MPEG-DASH by 29% in PSNR but faltering under volatile networks. In contrast, deadline-centric models like SQP [7] treat latency as sacrosanct, delivering 71% of frames within 100 ms—ideal for AR/VR—but crumbling under cross-traffic spikes [11]. Meanwhile, HTTP adaptive streaming saw reinvention via sliding-window LSTM models [8], which boosted bitrate consistency by 15% but ignored mobile energy constraints. These efforts, while groundbreaking, share a fatal flaw: protocols are designed in silos, assuming ideal networks and homogeneous user behavior. RFC 8867 [11]'s benchmarking exposed this, revealing that SQP's latency gains vanish when cross-traffic exceeds 50% capacity, while ASViS's layer prioritization falters under NOMA's imperfect SIC [10]. The rise of in-network computing [14] further complicates this landscape, redistributing adaptation logic to ISP middleboxes but clashing with edge-centric paradigms [1]. A harmonized protocol stack—blending deadline awareness, interference resilience, and distributed adaptation—remains elusive.

2.3 QoE's Identity Crisis: Perception vs. Precision

QoE optimization has fractured into two irreconcilable camps: predictive precision and reactive pragmatism. GRU-based models [6] epitomize the former, forecasting bandwidth 5–10 seconds ahead to preempt stalls with 89% accuracy. Yet, their reliance on historical patterns renders them blind to sudden drops—a weakness exploited by buffer-centric reactive strategies [9], which prioritize chunk deadlines over bitrate maximization, stabilizing QoE at the cost of resolution. Theoretical frameworks [12], [15] attempted mediation, showing users prefer steady 720p over volatile 4K (25% higher ratings) and that missing deadlines by >200ms increases abandonment by 30%. However, real-world RL deployments such as [5] reveal a fundamental divergence between theoretical QoE metrics and actual user preferences. In this context, 'perceived QoE' is defined as the subjective quality experienced by users—quantified through behavioral signals such as watch time, thumbs-up feedback, and skip events. Trained on over 30 million user sessions, the RL agents in [5] optimized these perceived signals, prioritizing high-resolution retention during emotionally significant scenes (e.g., climaxes, goal moments), even at the cost of higher stall probability. This behavior contradicts traditional metrics (e.g., average bitrate or rebuffering duration), revealing that human-centric engagement often outweighs purely technical optimization.

2.4 Edge-NOMA: The Scalability Mirage

Edge computing and NOMA promised a utopia of scalable multi-user streaming but collided with real-world imperfections. ECAS-ML [1] showcased edge-assisted ML's potential, yet its centralized architecture created bottlenecks, while NOMA-edge systems [10] boosted spectral efficiency by 20% but crumbled under imperfect SIC—residual interference slashed QoE by 12% in stadium-like density. In-network computing [14] emerged as a compromise, offloading logic to middleboxes to reduce client overhead by 38%, but at the cost of backhaul latency and API standardization gaps. These efforts, while innovative, ignored a critical reality: edge and NOMA are symbiotic, not siloed. For instance, NOMA's overlapping resource blocks demand dynamic bitrate-codebook coordination unaddressed by existing protocols [3], while edge caching's transcoding overhead negates ECAS-ML's latency gains. Theoretical models [12], [15] further failed to bridge this gap, lacking NOMA-aware benchmarks or hybrid latency models.

2.5 The Road Ahead: A Call for Cross-Layer Consilience

This survey's analysis culminates in three imperatives for the 6G era:

Unified MARL Architectures: Merge multi-agent bitrate fairness [2], deadline prioritization [7], and NOMA interference constraints [10] into a single reward framework.

Hybrid QoE Metrics: Bridge perceptual consistency [15], deadline adherence [12], and engagement signals [5] into a dynamic index.

In-Network NOMA Standardization: Develop middlebox APIs [14] to align bitrates with real-time SIC conditions [10], enforced via industry-wide benchmarks [11].

By transcending isolated innovations, the community can usher in an era where AI's adaptability, protocol-layer pragmatism, and human-centric design coalesce into self-orchestrating streaming ecosystems. The clock is ticking—6G's terahertz channels and AI-native air interfaces demand nothing less.

The preceding synthesis captures the macro-level trajectory of adaptive streaming evolution. In the following sections, we transition to a micro-level technical analysis—dissecting the architectural choices, optimization strategies, and empirical trade-offs across each innovation cluster. This deep dive enables a

cross-sectional understanding of where these methods stand today and how they can be bridged in future streaming ecosystems.

2. AI-Based Adaptation and Congestion Control: Architectures, Trade-offs, and Hybridization

AI-driven approaches dominate adaptive streaming research, leveraging edge computing, multi-agent systems, and predictive models. This section dissects five sub-categories—edge-assisted ML, multi-agent RL, DRL for TCP control, GRU-based prediction, and real-world RL—synthesizing insights from [1], [2], [4]–[6], [10], [13]. A comparative summary is provided in Table I.

2.1 Edge-Assisted ML:

ECAS-ML [1] integrates edge computing with supervised learning to predict network states, reducing rebuffering by 23% compared to client-only ABR algorithms like BOLA [3]. By offloading ML inference to edge servers, it minimizes end-to-end latency (critical for live streaming) but introduces centralization risks. Single-edge dependency creates bottlenecks during peak loads, a limitation unaddressed in [1], contrasting with [14]’s decentralized in-network computing. While achieving 12–28% QoE gains in simulated 4G environments, its reliance on offline parameter tuning limits adaptability to dynamic network conditions. For example, edge servers trained on daytime traffic patterns fail to adapt to evening congestion spikes, highlighting the need for real-time model updates.

2.2 Multi-Agent RL (MARL):

[2] employs MARL with expert guidance (e.g., buffer thresholds, throughput thresholds) to coordinate bitrate selection across short video segments. Expert rules reduce exploration time, achieving 18% higher QoE than solo RL agents by enforcing stable buffer levels (e.g., maintaining $\geq 5s$ buffer). However, MARL’s scalability in multi-user NOMA systems [10] remains untested, as overlapping resource blocks in NOMA complicate agent coordination. The framework achieves 53% utility gains with 2× fewer stalls but requires expert policies and is validated only on short video segments, neglecting long-form content’s variable engagement dynamics.

2.3 DRL for TCP Congestion Control:

[4] applies deep reinforcement learning (DRL) to TCP congestion control, dynamically adjusting congestion windows using real-time RTT and packet loss metrics. This reduces retransmission timeouts by 34% compared to Cubic TCP in congested networks [7] and achieves 11.7–87.5% latency reduction in Mininet emulations. However, during traffic bursts ($>2\%$ packet loss variance), DRL’s lack of explicit constraints leads to erratic window adjustments, causing jitter. Hybrid heuristic-RL policies [13] address this by enforcing conservative window sizes (e.g., ≥ 4 MSS during instability), ensuring baseline throughput at the cost of exploration flexibility.

2.4 GRU-Based Predictive Adaptation:

While predictive models like GRU-based bandwidth forecasting [6] demonstrate tangible gains in stall reduction, their integration within edge-NOMA ecosystems remains fragmented. For instance, although [6] reports 41% fewer stalls via temporal forecasting, its lack of cross-layer coordination renders it ineffective when interference from imperfect SIC arises, as noted in [10]. This underscores a missed opportunity: predictive models must be re-engineered to operate under interference-aware edge conditions—an integration yet unexplored across the referenced works.

2.5 Real-World RL Deployment: Bridging Simulation and Practice

[5] trains RL agents on live user feedback (e.g., thumbs-up/down signals, session duration, and interaction rates), optimizing for subjective QoE rather than technical metrics like bitrate. This approach achieves 15% higher perceived quality than simulation-based RL across 30M+ real-world sessions, prioritizing viewer retention through context-aware adaptations. For instance, agents learn to avoid resolution drops during critical video moments (e.g., climax scenes) to sustain engagement. However, regional variations in user behavior—such as mobile-heavy regions favoring lower bitrates versus Wi-Fi users tolerating higher resolutions—create non-stationary training data, necessitating frequent model updates to maintain accuracy. While validated at production scale, platform-specific optimizations (e.g., tightly coupled CDN integrations) complicate reproducibility [11], limiting its generalizability to diverse streaming ecosystems.

Approach	Key Innovation	Performance Gains	Strengths	Weaknesses
ECAS-ML [1]	Edge-assisted supervised learning	23% latency reduction, 18% fewer stalls	Edge offloading, live streaming-ready	Centralized bottlenecks, offline tuning
MARL [2]	Expert-guided multi-agent RL	53% utility gain, 2× fewer stalls	Fast inference, short-video optimized	Untested in NOMA, expert dependency
DRL-TCP [4]	PPO-based TCP window tuning	34% timeout reduction, 87.5% latency drop	Adaptive to congestion	Bursty traffic instability, high overhead
Real-World RL [5]	Bayesian-optimized RL deployment	15% higher perceived QoE (30M+ sessions)	Production-scale proven	Platform-specific, modest gains
GRU-ABR [6]	GRU bandwidth prediction	41% stall reduction, 89% prediction accuracy	Lightweight, mobility-ready	Fails under sudden drops

3. Protocols and Congestion Control Algorithms: Latency, Scalability, and Standardization

Protocol innovations prioritize low-latency streaming and QoE stability but face scalability and realism gaps. This section analyzes four approaches—SVC-aware ABR, deadline-centric congestion control, dynamic bitrate adaptation, and in-network computing—integrating insights from [3], [7], [8], [11], [12], [14]. A comparative summary is provided in Table II .

3.1 Advanced ABR Protocols: SVC-Aware Adaptation

ASViS [3] is an application-layer ABR strategy designed to work atop protocols like MPEG-DASH. It integrates scalable video coding (SVC) awareness into its logic, dynamically prioritizing base and enhancement layers according to buffer health, perceptual relevance, and throughput predictions. While not a transport-layer protocol per se, ASViS redefines adaptation behavior through a codec-aware decision engine. It outperforms MPEG-DASH by 29% in PSNR and achieves 5.8% higher VMAF by selectively dropping non-critical layers during congestion. The protocol maintains a minimal playback buffer (0.45 s) with zero rebuffering in simulations, leveraging predictive delivery modeling to optimize network utilization. For example, during bandwidth drops, ASViS discards enhancement layers (e.g., texture details) while preserving base layers (e.g., motion vectors), ensuring continuous playback. However, its reliance on stable throughput estimations conflicts with [8]’s dynamic bitrate analysis, which employs sliding-window HTTP adaptation to handle volatility, achieving 15% higher bitrate consistency. ASViS’s dependency on SVC codecs and lack of real-world deployment limit its applicability to heterogeneous streaming ecosystems, where hybrid codecs like AV1 dominate.

3.2 Low-Latency Congestion Control: Deadline Prioritization

SQP [7] redesigns congestion control around frame deadlines, reducing 95th percentile latency by 52% compared to Google’s GCC. By pacing frames to align with network capacity, it achieves 78% Wi-Fi utilization and delivers 71% of frames within 100 ms in real-world tests. For instance, in interactive AR/VR streaming, SQP prioritizes head movement frames over background updates, minimizing motion-to-photon delays. However, RFC 8867 [11]’s benchmarking exposes vulnerabilities under cross-traffic exceeding 50% of bottleneck capacity, where queuing delays spike by 4–8 ms, disrupting frame deadlines. While validated on AR/VR platforms, SQP’s reliance on precise clock synchronization (e.g., sub-millisecond accuracy) and untested performance in 5G/Beyond-5G networks raise scalability concerns, particularly in high-mobility scenarios.

3.3 Dynamic Bitrate Adaptation: Mitigating HTTP Overhead

[8] proposes a dynamic bitrate analysis technique using sliding-window throughput estimation, reducing HTTP overhead and achieving 15% higher bitrate consistency than BOLA [3]. The method employs an attention-LSTM model to forecast bandwidth, yielding 37% higher PSNR and 11.4% SSIM gains over buffer-based approaches in 4G networks. For example, during sudden bandwidth fluctuations, the LSTM predicts throughput trends 5–10 seconds ahead, enabling proactive bitrate switches (e.g., 720p → 480p) to avoid stalls. However, its focus on perceptual quality overlooks latency optimization—critical for live

sports or gaming—and computational overhead from ML inference (e.g., 1.2W power draw on mobile GPUs) restricts deployment on resource-constrained devices.

3.4 In-Network Computing: Redistributing Adaptation Logic

[14] offloads ABR logic to ISP middleboxes, reducing client overhead by 38% and enabling interference-aware bitrate adaptation in NOMA systems [10]. Middleboxes analyze aggregated traffic to allocate bitrates dynamically—e.g., prioritizing 4K streams for premium users during peak hours. While this approach minimizes backbone traffic, it conflicts with edge-centric models like ECAS-ML [1], which prioritize proximity over centralized control. For instance, edge servers process bitrate decisions locally, whereas middleboxes introduce 10–15ms latency for upstream coordination. The lack of standardized APIs for middlebox integration (e.g., OpenFlow or P4 support) further complicates large-scale adoption, leaving interoperability to proprietary solutions.

Protocol	Key Innovation	Performance Gains	Strengths	Weaknesses
ASViS [3]	SVC-aware layer prioritization	29% PSNR gain, 0.45 s buffer	Zero rebuffering, high perceptual quality	SVC dependency, simulation-only validation
SQP [7]	Deadline-paced congestion control	52% latency reduction, 78% Wi-Fi utilization	Ultra-low latency, real-world deployment	Bursty traffic sensitivity, 5G untested
Dynamic Bitrate [8]	LSTM-driven HTTP adaptation	37% PSNR gain, 15% bitrate consistency	Bandwidth-aware, robust under 4G volatility	Mobile incompatibility, latency unoptimized
In-Network [14]	Middlebox offloading for ABR logic	38% client overhead reduction	Scalable for NOMA, backbone efficiency	API standardization gaps, edge conflicts

Note: Unless otherwise stated, performance metrics are relative to BOLA [3] or MPEG-DASH baselines

4. Buffering and QoE Optimization: Predictive, Reactive, and Hybrid Paradigms

QoE optimization in adaptive streaming demands balancing buffer stability, playback continuity, and perceptual quality. This section dissects three strategies—predictive, reactive, and hybrid human-centric models—leveraging insights from [3], [5], [6], [9], [12], [15]. A comparative summary of predictive and reactive approaches is provided in Table III.

4.1 Predictive Mechanisms: GRU-Based Bandwidth Forecasting

GRU-based prediction [6] employs gated recurrent units to analyze temporal network patterns, forecasting bandwidth 5–10 seconds ahead with 89% accuracy. By preemptively adjusting bitrates, it reduces rebuffering by 41% compared to BOLA [3], which relies on instantaneous throughput. For

example, in mobile networks with periodic congestion (e.g., daily commute patterns), GRU models anticipate throughput dips, enabling seamless bitrate switches (e.g., 1080p → 720p) before stalls occur. However, during sudden bandwidth drops (e.g., tunnel handoffs), historical data becomes irrelevant, leading to 12–15% prediction errors that degrade QoE. This limitation is partially mitigated by edge offloading [1], but computational overhead (e.g., 1.8W GPU usage) restricts mobile deployment.

4.2 Reactive Strategies: Buffer-Centric Adaptation

[9]’s buffer-adaptive algorithm prioritizes chunk completion deadlines over bitrate maximization during network dips, achieving 22% higher QoE stability than predictive methods [6]. By dynamically scaling bitrates based on remaining download time (e.g., switching to 480p if <5s remains per chunk), it maintains playback continuity under volatility. However, this sacrifices average resolution (1080p → 720p), conflicting with [15]’s finding that users penalize resolution drops 25% more harshly than occasional stalls. While lightweight and compatible with standard HTTP streaming, the reactive approach lacks predictive insight, causing 2–3% more rebuffering in highly dynamic environments compared to GRU-based methods.

4.3 Hybrid Human-Centric Models

Theoretical frameworks bridge predictive and reactive paradigms by aligning technical metrics with human perception. [12]’s deadline-aware model quantifies user frustration thresholds, showing that missing playback deadlines by >200ms increases abandonment rates by 30%. This validates SQP’s [7] deadline prioritization but clashes with MARL’s [2] bitrate fairness focus. [5]’s real-world RL further hybridizes these concepts by training on live user feedback (e.g., thumbs-up/down signals), achieving 15% higher perceived QoE than simulation-based RL. For instance, agents learn to prioritize resolution during key scenes (e.g., sports highlights) while ensuring deadlines for live events. However, non-stationary real-world data (e.g., regional viewing habits) complicates reproducibility [11], and platform-specific optimizations hinder cross-service deployment.

4.4 Trade-offs and Unresolved Challenges

Predictive vs. Reactive

GRU-based prediction [6] reduces rebuffering by 41% in stable networks by forecasting bandwidth 5–10 seconds ahead with 89% accuracy, enabling proactive bitrate adjustments (e.g., 1080p → 720p before congestion). For instance, in predictable environments like daily commute routes with recurring 4G/LTE patterns, GRU models leverage historical trends to preempt stalls. However, during sudden drops—such as mobile handoffs between cell towers or tunnel entries—historical data becomes obsolete, causing 12–15% prediction errors that misalign bitrates and degrade QoE. In contrast, reactive methods like [9]’s buffer-adaptive algorithm prioritize chunk deadlines during volatility, dynamically lowering bitrates (e.g., 720p → 480p) to ensure playback continuity. This stabilizes QoE by 22% but sacrifices resolution consistency, as users endure 1080p → 720p transitions during 30% of streaming sessions. While GRU excels in stable networks, its computational overhead (e.g., 1.8W GPU usage) limits mobile use, whereas reactive approaches lack foresight, reacting only after quality degrades.

Human-Centric Gaps

Users overwhelmingly prefer steady 720p streaming over sporadic 4K bursts, rating consistent quality 25% higher despite 18% lower bitrate [15]. This preference stems from frustration with frequent resolution fluctuations, which disrupt immersion more than occasional stalls. However, deadline-centric protocols like SQP [7] prioritize latency, delivering 71% of frames within 100 ms but ignoring resolution stability. For example, SQP ensures live sports streams meet real-time deadlines but allows resolution drops during critical plays, contradicting viewer expectations. Meanwhile, codec-aware protocols like ASViS [3] optimize PSNR and VMAF but overlook subjective deadlines, failing to align technical metrics with human tolerance thresholds. Bridging this gap requires hybrid models that balance deadline adherence (e.g., <200ms delays [12]) with perceptual consistency (e.g., minimal resolution drops [15]), yet no existing framework reconciles these competing objectives.

Hybrid Potential

Integrating [5]’s real-world RL—trained on 30M+ user sessions—with [9]’s buffer-centric adaptation could harmonize stability and quality. For instance, RL agents could learn to prioritize resolution during high-engagement scenes (e.g., movie climaxes) using thumbs-up/down signals, while buffer rules enforce deadlines for time-sensitive content (e.g., live news). This synergy would allow proactive bitrate switches during predictable congestion (via GRU [6]) and reactive fallbacks during sudden drops (via [9]), minimizing rebuffering below 1% while preserving HD quality. However, challenges persist: non-stationary user behavior (e.g., regional viewing habits) complicates RL training, and conflicting objectives (e.g., bitrate vs. latency) demand multi-agent coordination. Current works like [5] and [9] operate in isolation, leaving a critical gap in human-aware hybrid systems. Addressing this requires cross-layer optimization frameworks that unify predictive ML, reactive thresholds, and subjective QoE metrics—an open research frontier.

Criterion	Predictive (GRU-ABR [6])	Reactive (Buffer-Adaptive [9])
QoE Gains	28% higher avg. bitrate, <0.5% rebuffering	17% smoother playback, <2% rebuffering
Bandwidth Prediction	GRU-based (89% accuracy)	None; relies on emergency fallback logic
Latency Handling	Proactive adjustments	Reactive to sudden drops
Strengths	High bitrate stability, learned patterns	Robust under abrupt network collapse
Weaknesses	GRU model dependency, mobile overhead	Resolution sacrifices, post-drop reactions
Deployment	Requires ML integration	Lightweight, HTTP-compatible

See Table I note for baseline comparison context

5. Edge Computing and NOMA-Based Solutions: Scalability vs. Practical Imperfections

Edge computing and Non-Orthogonal Multiple Access (NOMA) promise scalable multi-user streaming but face deployment hurdles rooted in spectral efficiency vs. realism, latency-complexity trade-offs, and standardization gaps. This section analyzes architectures, interference challenges, and theoretical critiques from [1], [10], [14], synthesizing their trade-offs in Table IV.

5.1 Edge Caching Architectures

ECAS-ML [1] offloads ML-based bitrate adaptation to edge servers, reducing client-side latency by 23% through proximity-based inference. By leveraging Long Short-Term Memory (LSTM) models trained on real 4G traces, it optimizes bitrate decisions (e.g., ~2772 kbps average) while minimizing stalls (16 per session) and quality switches, achieving an ITU-T P.1203 QoE score of 3.65—the highest among evaluated methods. However, its centralized edge nodes create single points of failure during peak loads and lack integrated caching, forcing repeated transcoding for popular content. This contrasts with [14]’s in-network computing, which distributes adaptation logic across ISP middleboxes, reducing client overhead by 38% but introducing 50–100 ms backhaul delays due to upstream coordination. The dependency on Radio Network Information Service (RNIS) integration further limits interoperability, as few commercial edge platforms support real-time data aggregation.

5.2 NOMA Integration: Spectral Efficiency vs. SIC Imperfections

[10] integrates adaptive streaming with NOMA edge caching, achieving 20% higher spectral efficiency than Orthogonal Multiple Access (OMA) systems by dynamically optimizing bitrates under power constraints. Through joint Lyapunov-Newton optimization, it delivers 75% higher bitrates and 27% smoother playback than baselines. However, these gains hinge on synthetic channel models and static user assumptions, ignoring real-world mobility (e.g., users moving between cells) and imperfect Successive Interference Cancellation (SIC). Residual interference degrades QoE by 12% in dense networks (e.g., stadiums with 10+ users per cluster), exposing a critical gap in NOMA-aware ABR protocols. Existing methods like ASViS [3] neglect interference-aware bitrate coordination, while RFC 8867 [11]’s congestion control benchmarks lack NOMA-specific test cases (e.g., interference spikes), undermining real-world validity.

5.3 Edge-NOMA Synergies and Conflicts

Scalability: NOMA [10] supports 50% more users per edge node than OMA by assigning overlapping resource blocks. However, dynamic bitrate-codebook coordination (e.g., adjusting QAM levels based on interference) remains unaddressed in current protocols like ASViS [3], leading to overestimated throughput during SIC failures.

Latency-Complexity Trade-off: ECAS-ML [1] reduces client latency by 23% but increases backhaul traffic by 18% due to centralized edge processing. In-network computing [14] alleviates backbone load but adds coordination overhead between middleboxes and edge nodes, complicating real-time adaptation.

Standardization Gaps: The absence of hybrid latency models [12] that account for edge-induced backhaul delays (e.g., 50–100 ms) invalidates assumptions in deadline-aware protocols. Meanwhile, NOMA-Edge [10]’s dependency on base station-integrated caching (gNB) and ECAS-ML’s RNIS requirements highlight interoperability challenges.

Criterion	<i>NOMA-Edge [10]</i>	<i>ECAS-ML [1]</i>
Core Approach	Joint bitrate-power optimization with SIC	LSTM-based bitrate adaptation at edge
QoE Objectives	Maximize bitrate/smoothness under interference	Optimize bitrate, stalls, and quality switches
Edge Utilization	Caching, transcoding, and power allocation	ML inference and ABR decision-making
Scalability	50% more users than OMA; static assumptions	Limited by backhaul traffic and client load
Strengths	High spectral efficiency, low-complexity	Real-world validation, low client latency
Weaknesses	Synthetic data, SIC dependency	Centralized edge nodes, no caching support

See Table I note for baseline comparison context

6. Theoretical Foundations and Modeling Approaches: Bridging Simulation and Reality

Theoretical works [11]–[15] provide frameworks to validate experimental systems, exposing gaps in realism and scalability.

6.1 Benchmarking Frameworks

SQP’s Vulnerabilities:

RFC 8867 [11] reveals SQP [7] fails to maintain sub-200ms latency when cross-traffic exceeds 50% of bottleneck capacity. For example, in a 10 Mbps link, SQP’s deadline prioritization breaks down under 5 Mbps competing traffic, causing 40% of frames to miss deadlines.

DRL-Based TCP Instability:

[4]’s DRL approach reduces retransmission timeouts by 34% in stable congestion but becomes unstable under bursty traffic with >2% packet loss variance. Hybrid heuristic-RL [13] addresses this by enforcing minimum congestion window sizes, reducing timeout variability by 22%.

6.2 Hybrid Heuristic-RL Models

[13] combines rule-based safety checks (e.g., minimum buffer thresholds) with MARL exploration, reducing [2]’s training time by 45% while retaining 92% of its QoE gains. This hybrid approach is validated in [12]’s deadline-aware framework, which shows that heuristic safeguards reduce deadline misses by 33% in live deployments.

6.3 Perceptual Quality vs. Technical Metrics

[15]’s perceptual model prioritizes consistent resolution (e.g., steady 720p) over sporadic 4K bursts, demonstrating that users rate steady streams 25% higher despite 18% lower bitrate. This directly conflicts with [3]’s ASViS protocol, which maximizes PSNR through SVC layer switching, and [2]’s MARL, which optimizes for bitrate fairness.

7. Comparative Analysis: Cross-Domain Evaluation of All 15 Papers

Table 1: Holistic Comparison of Research Contributions

Category	Key Papers	Strengths	Limitations	Theoretical Support
AI/ML Adaptation	[1], [2], [4]–[6]	Proactive adaptation, latency reduction	High compute overhead, scalability issues	RFC 8867 [11], Hybrid [13]
Protocols & Congestion	[3], [7], [8], [14]	Ultra-low latency, QoE stability	Assumes ideal networks, lacks NOMA support	Deadline-aware [12], In-network [14]
QoE Optimization	[6], [9], [12], [15]	Human-centric metrics, buffer stability	Reactive methods lack proactivity	Perceptual [15], Deadline [12]
Edge/NOMA	[1], [10], [14]	Spectral efficiency, scalability	Imperfect SIC, edge dependency	In-network [14]
Theoretical Models	[11]–[15]	Real-world benchmarks, hybrid policy validation	Limited integration with edge/NOMA	N/A

Critical Insights from Cross-Comparison

AI-Protocol Synergy Gap:

MARL [2]’s multi-agent bitrate fairness could complement SQP [7]’s deadline prioritization. For instance, MARL agents could negotiate bitrates on parallel paths while SQP ensures frame deadlines, but no work explores this integration.

NOMA’s Unresolved Challenges:

[10]’s NOMA edge caching suffers from imperfect SIC, which introduces residual interference equivalent to 3–5 dB SNR loss. Current ABR methods [3], [8] lack interference-aware bitrate adaptation, leading to overestimated throughput and rebuffering.

Human-Centric Divergence:

[15]’s user studies show a 25% preference for steady 720p over volatile 4K, conflicting with ASViS [3]’s PSNR-driven layer switching and MARL [2]’s bitrate fairness. A unified metric combining resolution consistency and deadline adherence is needed.

8. Conclusion and Future Directions: Bridging Gaps Across Layers

Based on the synthesis of [1]–[15], this survey proposes three transformative pathways to unify AI-driven adaptation, protocol-layer innovations, and human-centric QoE. Our analysis transcends mere aggregation of trends—it advocates for cross-disciplinary integration and metric standardization, grounded in the limitations and contradictions exposed by the reviewed works. Below, we refine our recommendations with actionable evaluation frameworks, implementation roadmaps, and a timeline for adoption.

1. Multi-Objective MARL: A Unified Reward Framework

We advocate a shift toward MARL architectures that harmonize bitrate fairness, latency, and physical-layer constraints. To operationalize this:

Reward Function Prototype:

R

$=$

0.5

.

Bitrate Fairness

⌋

[2]

+

0.3

.

Deadline Adherence (1 - Missed Deadlines)

⌋

[7], [12]

–

0.2

.

SIC Failures

⌋

[10]

R=

[2]

0.5·Bitrate Fairness

+

[7], [12]

0.3·Deadline Adherence (1 - Missed Deadlines)

-

[10]

0.2·SIC Failures

Bitrate Fairness: Normalized Jain’s index across users.

Deadline Adherence: Percentage of frames delivered within 200ms [12].

SIC Failures: Ratio of packets corrupted by residual interference [10].

Validation: Test in a NOMA-edge simulator combining [2]’s MARL, [10]’s interference model, and [7]’s congestion control.

Contribution Highlight: This framework directly addresses the AI-protocol silo gap identified in Section 7, offering the first MARL design to incorporate physical-layer impairments.

2. Perceptual-Deadline Hybrids: A Dynamic QoE Index

We propose a Hybrid QoE Index (HQoI) to unify subjective and technical metrics, resolving the conflict between [12] and [15]:

HQoI

=

0.4

·

Resolution Consistency

⊂

[15]

+

0.3

.

Deadline Compliance

⊃

[12]

+

0.3

.

Rebuffering Ratio

⊃

[6], [9]

HQoI=

[15]

0.4·Resolution Consistency

⊃

[12]

+

+

[12]

0.3·Deadline Compliance

⊃

[12]

+

+

[6], [9]

0.3·Rebuffering Ratio

Resolution Consistency: Percentage of time resolution $\geq 720p$.

Deadline Compliance: Chunks delivered within 200ms thresholds.

Rebuffering Ratio: Total stall duration per session.

Implementation Roadmap:

Short-Term (6–12 months): Integrate HQoI into existing ABR algorithms (e.g., ASViS [3], GRU [6]) via software updates.

Long-Term (2–3 years): Embed HQoI in standardization bodies (e.g., DASH-IF) as a mandatory QoE metric.

Contribution Highlight: This index formalizes the trade-offs between human perception and protocol efficiency—a gap no prior work [3], [12], [15] systematically addresses.

3. In-Network NOMA: Middlebox-Driven Codebook Adaptation

We urge adoption of middlebox logic [14] to mitigate imperfect SIC in NOMA [10], aligning bitrates with real-time interference levels.

Codebook-Bitrate Mapping Table:

SIC SNR (dB)	Max Sustainable Bitrate	Codebook
≥ 25	8 Mbps	64-QAM
15–25	4 Mbps	16-QAM
< 15	2 Mbps	QPSK

Deployment Timeline:

Year 1: Lab testing with simulated NOMA interference [10].

Year 2: ISP trials using [14]'s middlebox APIs.

Year 3: Commercial rollout in 5G-Advanced networks.

Contribution Highlight: This is the first proposal to unify NOMA's physical layer with in-network adaptation, directly tackling the "ideal SIC assumption" critique of [10].

4. Cross-Layer Benchmarking: A Call for Standardization

We demand industry-academia collaboration to develop unified simulators and datasets, addressing the fragmented evaluation critiqued in Sections 6–7.

Benchmark Components:

Network Traces: Real-world NOMA interference patterns [10], edge backhaul delays [14], and bursty cross-traffic [11].

QoE Metrics: HQoI (above), spectral efficiency [10], and training stability scores [13].

Timeline:

2024: Release open-source simulators integrating [1], [7], [10].

2025: Standardize benchmarks via IETF/RFC extensions [11].

5. Timeline Vision: From Theory to Deployment

Based on technical readiness and industry inertia, we project:

Immediate (0–1 year):

Perceptual-deadline hybrids [12], [15] deployable via CDN firmware updates.

MARL-NOMA prototypes [2], [10] tested in lab environments.

Mid-Term (1–3 years):

In-network NOMA [14] trialed by Tier-1 ISPs.

Hybrid QoE Index adopted by streaming platforms.

Long-Term (3–5 years):

Cross-layer simulators mandated in academic evaluations.

80% of live streams using AI-protocol hybrid systems.

Final Synthesis

This survey’s core contribution lies in its prescriptive roadmap—not merely cataloging gaps but providing actionable, quantifiable, and temporally grounded solutions. By redefining reward functions, standardizing metrics, and orchestrating cross-layer benchmarks, we bridge the chasm between AI’s potential and protocol-layer pragmatism. The timelines and frameworks proposed here are not speculative—they are engineering mandates for an industry poised to scale adaptive streaming into the 6G era.

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